



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (CBCS) Degree Examinations
 December 2024

MA2004-1 – VECTOR CALCULUS AND TRANSFORM TECHNIQUES
 (For EE, EC, ACT, VLSI)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, then $\text{div}\vec{r} = \underline{\hspace{2cm}}$ and $\text{curl}\vec{r} = \underline{\hspace{2cm}}$
 - $\hat{i} + \hat{j} + \hat{k}$ and 0
 - 3 and 0
 - 3 and $\vec{0}$
 - 0 and $\vec{0}$
- If $\phi = x^2 + y - z - 1$, then $|\nabla \phi|$ at $(1, 0, 0)$ is
 - $\sqrt{5}$
 - $\sqrt{2}$
 - $\sqrt{6}$
 - 1
- The value of $\text{curl}(\text{grad}f)$, where $f = 2x^2 - 3y^2 + 4z^2$ is
 - $\vec{0}$
 - $4x - 6y + 8z$
 - $4x\hat{i} - 6y\hat{j} + 8z\hat{k}$
 - 3
- A vector field which has a vanishing divergence is called as
 - Scalar field
 - Solenoidal vector field
 - Irrotational vector field
 - Rotational vector field
- Let ϕ be a scalar potential of the conservative vector field $\vec{F}$. Then the work done in moving a particle along a curve from point A to B is
 - $\phi(B)$
 - $\phi(B) - \phi(A)$
 - 0
 - $\phi(A)$
- Suppose S is an open, two-sided surface bounded by a simple closed curve C and $\vec{F}$ has continuous derivatives. Then by Stokes' theorem which of the following is true?
 - $\int_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot \hat{n} dS$
 - $\int_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \cdot \vec{F}) \cdot \hat{n} dS$
 - $\int_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) dS$
 - $\int_C \vec{F} \times d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot \hat{n} dS$
- Gauss divergence theorem relates
 - line and surface integral
 - surface and volume integral
 - line and volume integral
 - line integral, surface integral and volume integral
- The value of $\int_C x dy - y dx$ where C is $x^2 + y^2 = 1$ is
 - 0
 - $\frac{1}{2}$
 - 1
 - 2π
- If $f(z) = u(r, \theta) + iv(r, \theta)$ is analytic then $\frac{\partial u}{\partial r} =$
 - $\frac{\partial v}{\partial \theta}$
 - $\frac{1}{r} \frac{\partial v}{\partial \theta}$
 - $-r \frac{\partial v}{\partial \theta}$
 - $\frac{1}{r} \frac{\partial v}{\partial r}$

10. A point where a function ceases to be analytic is known as a
 A) Singular point B) Non – singular point
 C) Pole D) Residue
11. If $f(z) = x + ay + i(bx + cy)$ is analytic, then the values of a, b, c are
 A) $a = 1$ and $c = -b$ B) $b = 1$ and $a = -c$
 C) $a = b = c = 1$ D) $c = 1$ and $a = -b$
12. If $2x - x^2 + ay^2$ is harmonic, then the value of a is
 A) 1 B) 2
 C) 3 D) 0
13. According to Cauchy's integral formula, if $f(z)$ is analytic everywhere in a region R bounded by a closed curve C then for $a \in R$, $\oint_C \frac{f(z)}{z-a} dz$ is equal to
 A) $2\pi i$ B) $f(a)$
 C) $2\pi i f(a)$ D) 0
14. The residue of $f(z) = \frac{2z-1}{z(z+1)(z-3)}$ at $z = 0$ is
 A) 0 B) 2
 C) $1/3$ D) 5
15. For the function $f(z) = \frac{z^2+3}{z^2(z+4)}$ the pole $z = 0$ has order
 A) 1 B) 0
 C) 3 D) 2
16. The poles of $f(z) = \frac{z^2+1}{1-z^2}$
 A) 0 B) ± 1
 C) $\pm i$ D) 2
17. If $f(x) = x^2$ in $-\pi < x < \pi$, $f(x + 2\pi) = f(x)$ then the value of the Fourier coefficient b_n is
 A) 0 B) $\frac{2\pi^2}{3}$
 C) $\frac{\pi^2}{3}$ D) $\frac{2\pi}{3}$
18. The half range Fourier sine series expansion of $f(x)$ in $(0, 1)$ is
 A) $f(x) = \sum_{n=1}^{\infty} b_n \sin nx$ B) $f(x) = \sum_{n=1}^{\infty} b_n \sin n\pi x$
 C) $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} b_n \sin n\pi x$ D) $f(x) = \sum_{n=1}^{\infty} b_n \sin 2\pi x$
19. If $Z(u_n) = U(z)$ then $Z(nu_n)$ is equal to
 A) $zU(z)$ B) $-zU(z)$
 C) $z \frac{dU}{dz}$ D) $-z \frac{dU}{dz}$
20. If $u_n = \begin{cases} 5^{-n} & \text{if } n \geq 0 \\ 0 & \text{if } n < 0 \end{cases}$ then $Z(u_n)$ equals
 A) $\frac{z}{z-5}$ B) $\frac{z}{z+5}$
 C) $\frac{5z}{5z-1}$ D) $\frac{5z}{5z+1}$

PART – B: DESCRIPTIVE ANSWER QUESTIONS**Unit – I**

- | | Marks | BT* | CO* | PO* |
|--|-------|-----|-----|-----|
| 1. a) Show that the vector field $\vec{F} = 2xy\hat{i} + (x^2 + 2yz)\hat{j} + (y^2 + 1)\hat{k}$ is conservative. Also find its scalar potential. | 6 | L*1 | 2 | 2 |
| b) Prove that the cylindrical co-ordinate system is an orthogonal curvilinear co-ordinate system. | 6 | L2 | 1 | 1 |

- c) A particle moves along the curve $x = 2t^2, y = t^2 - 4t, z = 3t - 5$, where t is the time variable. Find the component of its velocity at time $t = 1$ in the direction given by $\hat{i} - 3\hat{j} + 2\hat{k}$. 4 L1 1 2
2. a) State and prove Green's theorem in the plane. 6 L2 2 1
- b) Using Gauss divergence theorem, evaluate $\iint_S \vec{F} \cdot \hat{n} dS$, where $\vec{F} = (y - z + 2)\hat{i} - (yz + 4)\hat{j} - xz\hat{k}$ and S is the closed surface of the cube bounded by $x = 0, x = 2, y = 0, y = 2, z = 0$ & $z = 2$. 6 L1 2 2
- c) Define an irrotational vector field. Determine the constants a, b, c so that the vector field $\vec{F} = (axy + z^3)\hat{i} + (bx^2 + z)\hat{j} + (bxz^2 + cy)\hat{k}$ is irrotational. 4 L1 1 2
3. a) Find the directional derivative of $\phi = 2xy + z^2$ at $(1, -1, 3)$ in the direction of the vector $\hat{i} + 2\hat{j} + 2\hat{k}$. 6 L1 1 2
- b) With usual notation prove that $\int_C \vec{F} \cdot d\vec{r}$ is independent of the path of integration if and only if $\oint_C \vec{F} \cdot d\vec{r} = 0$, where C is a simple closed curve. 6 L2 2 1
- c) If $\phi = 2x^3y^2z^4$ and $\vec{F} = \nabla\phi$. Find $\nabla \cdot \vec{F}$ at $(1, -1, 1)$. 4 L1 1 2
- Unit – II**
4. a) Find the bilinear transformation which maps $z = 1, i, -1$ onto $w = 0, 1, \infty$. 6 L1 3 2
- b) If $f(z) = u(x, y) + iv(x, y)$ is analytic in a domain D , then prove that at any point of D , $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$. 6 L2 3 1
- c) Using Cauchy's integral formula, evaluate $\int_C \frac{e^{2z}}{(z-1)(z-2)} dz$, if C is the circle $|z| = 3$. 4 L1 4 2
5. a) State and prove Cauchy's integral theorem. 6 L2 4 1
- b) Using Residue theorem, evaluate $\int_C \frac{2z-1}{z(z+1)(z-3)} dz$, where C is the circle $|z| = 2$. 6 L1 4 2
- c) Show that the function $f(z) = e^z$ is analytic everywhere. 4 L1 3 2
6. a) Define a harmonic function. Prove that the real and imaginary parts of an analytic function are harmonic functions. 6 L2 3 1
- b) Obtain the Laurent's expansion of $f(z) = \frac{1}{(z+1)(z+3)}$ in the region (i) $1 < |z| < 3$ (ii) $|z| > 3$. 6 L1 4 2
- c) Determine the analytic function $f(z)$ whose real part is $u = y + e^x \cos y$. Also find $f'(z)$. 4 L1 3 2
- Unit – III**
7. a) Obtain the Fourier series of $f(x) = \begin{cases} 0 & , -2 < x < 0 \\ 1 & , 0 < x < 2 \end{cases}$ and $f(x+4) = f(x)$. 6 L2 5 2
- b) Find the Fourier series containing the first harmonics from the following table.
- | | | | | | | |
|-----------|-----|-----|-----|-----|-----|-----|
| x° | 0 | 60 | 120 | 180 | 240 | 300 |
| y | 7.9 | 7.2 | 3.6 | 0.5 | 0.9 | 6.8 |
- c) Obtain the Fourier series of $f(x) = x, -\pi < x < \pi$ and $f(x + 2\pi) = f(x)$. 4 L1 5 2

8. a) Find the bilateral Z-transform and region of convergence of

$$u_n = \begin{cases} 4^n & \text{if } n < 0 \\ 2^n & \text{if } n \geq 0 \end{cases}$$
- b) Using unilateral Z-transforms, solve $u_{n+2} + 4u_{n+1} + 3u_n = 3^n$ with $u_0 = 0, u_1 = 1$.
- c) State and prove convolution theorem for bilateral Z-transforms.

6	L2	5	2
6	L1	5	2
4	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
